Nate Phillips

Exploring Periodicity in Economic Data

For my final visualization project, my goal was to design a novel visualization and to investigate its effectiveness. Specifically, I designed a visualization to illustrate periodic trends. This visualization takes the normal line graph of an x/y data set and graphs it onto a circle, such that the radius of the circle represents the value of y, and the angular position represents the value of x. Using this method, it would be possible to include several of these radial graphs within one circle, allowing for easy, side-by-side comparison of data. The goal of this project is to use such a visualization to analyze the periodic trends for a given data set, in this case the CPI (consumer price index) values in the United States over the past sixty-odd years.

To determine the effectiveness of this circle visualization, I also included a normal line graph that can be split into several different line graphs, in much the same manner that the circle visualization can be split into several different levels. By comparing the relative effectiveness of these two visualizations, I hope to highlight any shortcomings the circle visualization may have and find additional avenues for improvement.

The primary goal for this visualization is to illustrate the periodicity of the data. Ideally, using this visualization will allow users to find patterns and trends in the data that they would not normally be able to find. Of course, this visualization also has a few secondary goals; it is important that the visualization be easy to use and present a clear picture of the data, in general. Without these two features, the usability of the visualization would suffer dramatically, and, as such, it would be much less useful.

I used D3.js, a javascript library, to produce this visualization and display it on the web. The visualization contains both the circle visualization and a normal line chart, with buttons for selecting which visualization to display on screen, the amount of data to display, and the number of partitions into which the data is split. Each data point on the visualization, if hovered over, will display the details of the data associated with it. This allows users to investigate the specific values at certain points, while still displaying the overall trends of the data.

In my search through the visualization literature, I was not able to find many examples of similar visualizations. One example I found that dealt with visualizing CPI data took a different, more simplistic approach [1]. It simply graphed each month as a bubble (without labeling the months, as far as I can tell) and then plotted the ten-year inflation rate vs. 2012's inflation rate. I found a couple of other papers that dealt with visualizing economic data, but, in both, most of the elements were unrelated. I know, from a previous visualization class, that graphs similar to my circle visualization have been previously used in visualization research, particularly in illustrating periodic trends. In one example that I was able to find, one group of researchers created a spiral graph to illustrate the patterns of sunlight and darkness over the course of several days [2]. This visualization, while notably different than my visualization, strongly resembles what I am attempting here. It uses more or less the same type of display that I am using; however, instead of using line graphs to represent the sunlight, this visualization uses color.

On examining the circle visualization and comparing it to the line charts, I found that, while the circle visualization is reasonably effective, it does suffer from some issues. Most notably, it tends to be significantly harder to detect minor changes or trends within the data in each subsection of the graph. This is likely caused by the angular difference between the line graphs and the circle graphs. In other words, it's a lot easier to see a small change in a line that is straight than it is to see a small change in a curve. Admittedly, it seems likely that the circle visualization would perform more effectively on data with more dramatic changes. Such data would produce a change in the line that would likely be noticeable even with the presence of the curve angle. Still, however, for detecting small changes this circle visualization does not seem to be particularly effective.

In another example, the circle visualization splits up the circle by giving each line graph an equal length of the radius. In other words, for each line graph, the y-axis will be the same size. However, due to the nature of circles, this means that as the graphs move closer to the outside of the circle, the x-axis, and thus the graphed line, will grow noticeably longer. This does have a noticeable effect on the visualization, in that the outer rings of information are much larger and much easier to see than the inner regions. While this is an important issue to consider, it is not likely to significantly interfere with the visualization's goals, since, for the outer rings, the trends will be the same, just somewhat enlarged.

Overall, then, I think that the line graphs are more effective at illustrating periodicity than the circle visualization. In part, this is not particularly surprising; line graphs are well-established visualization tools that most people can readily and easily understand, whereas the circle visualization is both less common and more complex. It may be that there is some way to tweak the circle visualization to make it more effective. For example, it might be beneficial to zoom in so that users could more easily analyze the changes and patterns in the data. However, such a change would also detract from the ability of users to get the full picture from the visualization. Thus, any such change would have both upsides and downsides. Regardless, the circle visualization, while not as generally effective as the line graph, does have the potential to be an effective visualization method.

References

[1] B. Meyer, “Visualizing Disinflation…And No, We’re Not There Yet,” Inflation and Price Statistics.

[2] M. Alexa, W. Müller, M. Weber, “Visualizing Time-Series on Spirals.”